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MASTER'S DEGREE – ENVIRONMENTAL AND BIODIVERSITY MANAGEMENT
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ANALYTICAL APPROACH TO THE GENUS *ANTHURIUM* IN MARTINIQUE: MORPHOMETRIC, TAXONOMIC AND NOMENCLATURAL ASPECTS

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CONSERVATOIRE BOTANIQUE NATIONAL MARTINIQUE



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1 SCIENTIFIC STUDY REPORT [6630 words]

1.1 INTRODUCTION [1757 words]

The genus *Anthurium* Schott of the monophyletic subfamily Pothoideae (Cabrera et al. 2008) currently includes 1363 species in 2025 (Bánki et al. 2024), making it the most diverse taxonomic group in the Araceae family, with a recent estimate of around 3000 species expected by scientists (Boyce et al. 2020). Strictly Neotropical, from Mexico to southern Brazil, northern Argentina, Paraguay, and the Caribbean basin, it is highly diverse (Carlsen et al. 2013). *Anthurium* species inhabit altitudes from 0 to 4000 meters, reaching frost-prone areas, mostly on Andean slopes, in Central America, and in the Chocó (Jørgensen et al. 1999). Diversity remains high in southwestern Mexico and in the Brazilian Atlantic Forest (Reimuth et al. 2020), with significant endemism at low to moderate altitudes, particularly in the Andes of Ecuador, Colombia, and Peru (Reimuth et al. 2020). The biological types are also highly diversified, with terrestrial species, rheophytes, lithophytes, lianas, epiphytes, and hemi-epiphytes (Croat 1988; Huggett et al. 2010; Steenis 1987).

This genus is characterized by bisexual tetramerous flowers with tepals, a free spathe, a homogeneous spadix, sympodial growth, alternate spiral phyllotaxy, a non-flattened petiole, an apical geniculum, and a reticulate minor venation network with collective veins along the margins (Grayum 1990; Mayo, Bogner, et al. 1997). *Anthurium* species are distinguished by the presence of two types of cataphylls: prophyll and mesophyll (Ray 1987). While the genus is easily identifiable by its reproductive structures, taxonomic studies have focused more on vegetative characters than reproductive ones due to the majority of studies being based on dried specimens, making observation challenging (Croat 1983; Grayum 1990; Mayo, Bogner, et al. 1997). Croat indicates that several vegetative characters are useful for describing and distinguishing species within the genus, including petiole section, leaf venation, cataphylls (their evolution on the stem and over time), and geniculum traits (Croat and Bunting 1979). The genus *Anthurium* is monophyletic, composed of 18 major sections and 2 series, bringing the total to 20 groups (Croat and Carlsen 2020; Carlsen et al. 2013; Carlsen et al. 2019). A recent molecular phylogeny shows that 8 groups are monophyletic, while the majority of the recognized sections are polyphyletic (Carlsen et al. 2013; Carlsen et al. 2019). Comparison of molecular results and traditional morphological characters used in classification revealed that only 7 sections form well-supported monophyletic groups, defined by a coherent set of morphological synapomorphies (Carlsen et al. 2019). In contrast, polyphyletic sections group species sharing homoplastic morphological characters (Carlsen et al. 2019).

Most genetic and descriptive work on the genus has been carried out in Central America, with currently 389 taxa (Vannini et al. 2022). Some revisions have included species from both Central and South America, such as those of the section *Stipitata* Croat & Delannay (Delannay et al. 2024) and the section *Porphyrochitonium* (Schott) Engl. (Ortiz et al. 2024). However, few recent revisions have specifically focused on South American sections and taxa. Historically, the section *Pachyneurium* (Schott) Engl. and the section *Leptanthurium* Schott ex Engl. have been revised (Mayo 1982; Croat 1991). The situation is similar in the Caribbean, where recent studies have mainly focused on *Anthurium* species with a "bird's nest" habit (Mayo 1982), as well as on the section *Dactylophyllum* (Schott) Engl., which includes *Anthurium palmatum* (L.) Schott (Croat and Carlsen 2013). Although the diversity of the genus in the Caribbean is still poorly documented due to the lack of comprehensive studies, available data suggest relatively low species diversity

(Royal Botanic Gardens, Kew 2025; Catalogue of Life 2025; Muséum national d'Histoire naturelle 2025). This island arc, the Caribbean basin, is considered one of the 34 biodiversity hotspots (Mittermeier et al. 2004). It is also one of the five hottest spots globally (Myers et al. 2000), though heavily impacted by high levels of extinction and degradation, leaving only 11.3% of primary vegetation (Myers et al. 2000; Maunder et al. 2008).

The Caribbean basin results from the subduction of the Atlantic plate beneath the Caribbean plate, forming two distinct geological segments: a Mesozoic base in the north, common to the Greater Antilles, and Eocene formations in the south (Bouysse et al. 1983; Andreieff et al. 1987; Germa 2008). Magmatism is intermittent, shifting from tholeiitic to calc-alkaline series, reflecting variations in volcanic activity (Wester-camp 1988). Martinique stands out for its active volcanism, structured in three phases of the Lesser Antilles arc: an ancient submarine arc (25 - 21 Ma), an intermediate arc (16 - 7 Ma), and a recent arc (5 - 7 Ma to the present), including volcanoes such as Morne Jacob, Pitons du Carbet, Mont Conil, and Montagne Pelée (Germa 2008; Labanieh 2009; Ueckmann et al. 2017).

The topography of Martinique is divided into five major units aligned northwest/southeast (Portecop 1979). The north is marked by high volcanic massifs (Montagne Pelée, Pitons du Carbet), while the south presents a moderate relief alternating hills and plains (Portecop 1979) (Figure 1). This configuration creates slopes exposed or sheltered from the trade winds, strongly influencing vegetation (Portecop 1979). Over 50% of the island has slopes greater than 20%, which promotes erosion and influences the evolution of the flora (Portecop 1979). The topographical heterogeneity resulting from the geological history generates various ecological niches based on altitude, temperature, and rainfall (Portecop 1979). Three altitudinal levels structure the vegetation: Lower Tropical (<250 m, >25°C), hot and dry, dominated by xerophytic and mesophytic series (*Tabebuia* spp., *Ceiba pentandra*, *Ficus citrifolia*); Upper Tropical (250–1000 m, 20–25°C), more humid, with hygrophilous forests (*Sloanea* spp., *Podocarpus coriaceus*, *Simaruba amara*); Mountain Tropical (>1000 m, <20°C), with montane vegetation (*Guzmania* spp., *Pitcairnia* spp., *Sciodaphyllum* spp.) on ridges or slopes (Portecop 1979). There are also coastal edaphic successions, such as mangroves (*Rhizophora mangle*, *Avicennia* spp.) and sandy beaches (*Thespesia populnea*) (Portecop 1979) (Figure 2). This classification reflects the interaction between climate, altitude, and soil in organizing vegetation.

In Martinique, *Anthurium* occupies a range of ecological niches, from xerophilous, mesophilous, and hygrophilous zones to altitudes above 1000 m, except along the coastline (Howard 1979)(Figure 2). These species exhibit a wide range of forms: acaulescent or caulescent, epiphytic, lithophytic, climbing, or semi-climbing (Howard 1979; Mayo 1982; Fournet 2002). The list of *Anthurium* species present in Martinique varies depending on the taxonomic references consulted. The taxa *Anthurium acaule* (Jacq.) Schott, *Anthurium cordatum* (L.) Schott, *Anthurium grandifolium* (Jacq.) Kunth, and *Anthurium palmatum* (L.) Schott are unanimously accepted as present on the island (Royal Botanic Gardens, Kew 2025; Catalogue of Life 2025; Muséum national d'Histoire naturelle 2025; Missouri Botanical Garden 2025), and in ITMA (the taxonomic reference of the CBNMq). Their presence has been recently confirmed in 2012 (Acevedo-Rodríguez et al. 2012) and remains confirmed to this day (CBNMq database). In contrast, *Anthurium andraeanum* Linden ex André, although frequently observed on the island, is not native (Liu et al. 2025). All references agree that it is a cultivated species, introduced for floricultural production in Martinique (Alvarez et al. 2006). The POWO, COL, and Tropicos references also indicate that *Anthurium willdenowii* Kunth is endemic to the Lesser Antilles, without specifying its presence or absence in Martinique, even though no

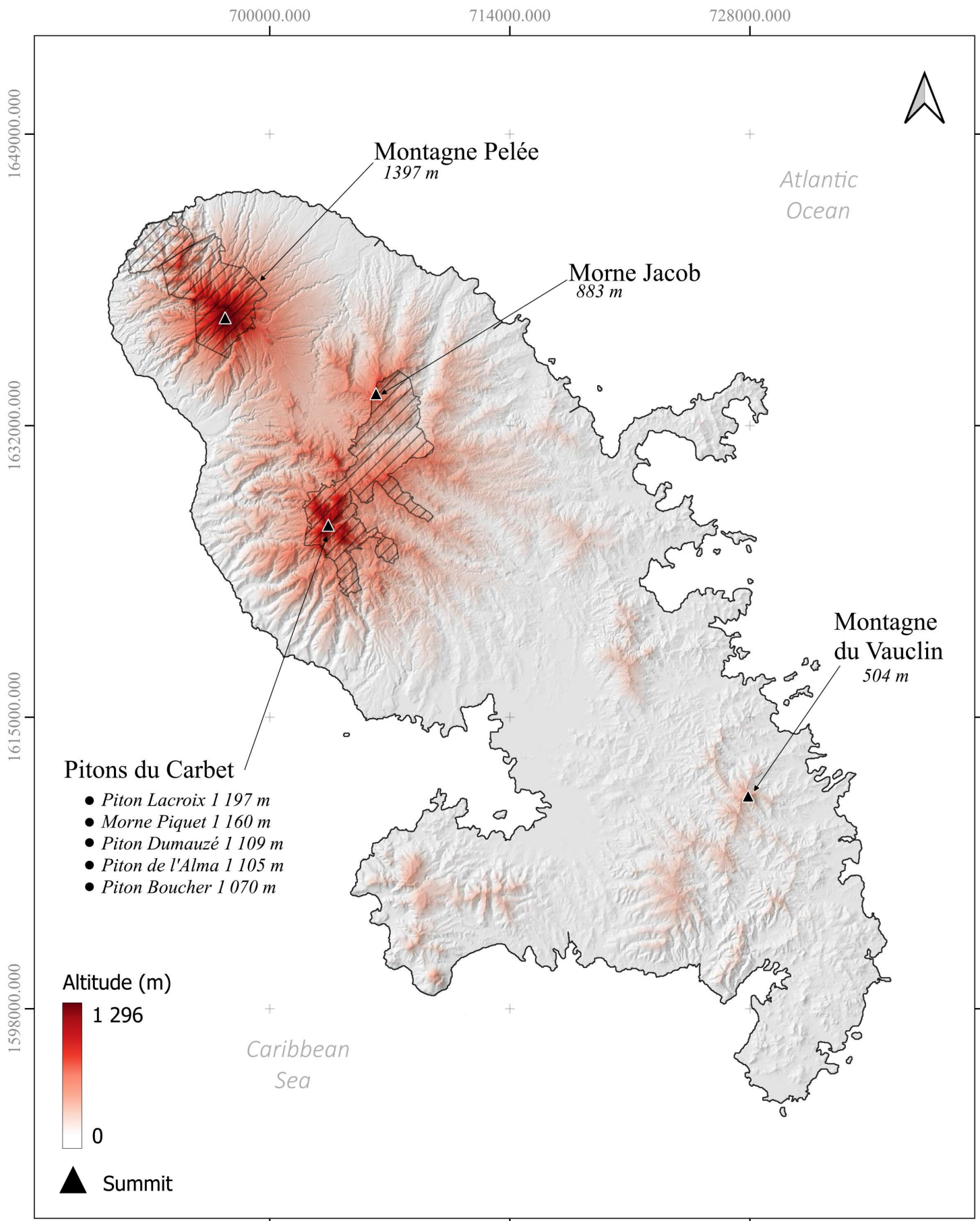


Figure 1: Altitudinal map of the island of Martinique, where a deep red color indicates higher elevation. Major peaks are marked with triangles, and hatched areas correspond to Integral Biological Reserves. Pixel size: 25 × 25 m. © Elian Vuillemot

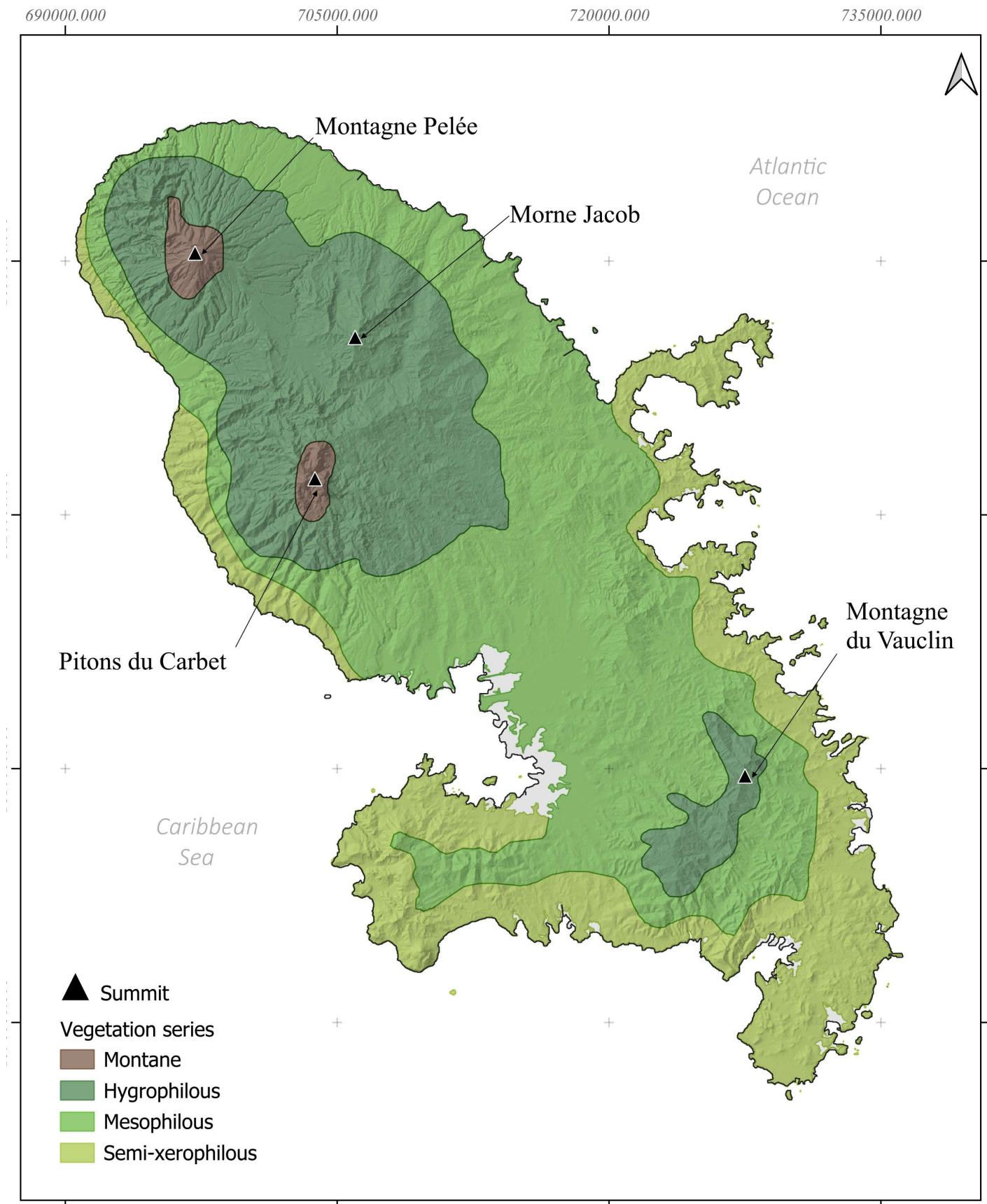


Figure 2: Vegetation series map of the island of Martinique: montane (brown), hygrophilous (dark green), mesophilous (green), and semi-xerophilous (yellow). Data based on Portecop (1979) and transcribed by CBNMq Portecop 1979. Major peaks are indicated by triangles. Pixel size: 25 × 25 m. © Elian Vuillemot.

observations have been made to date on the island and it appears to originate from Barbados (Howard 1979). Meanwhile, TaxRef and ITMA explicitly report its absence. Regarding *Anthurium dominicense* Schott, its presence is reported by POWO, COL, TaxRef, and ITMA, but no observations are mentioned on the island in recent databases. Furthermore, TaxRef, Tropicos, and ITMA have not updated their databases or have not accepted the recent synonymisation of *A. cordatum* (L.) Schott by *Anthurium guildingii* Schott, whereas POWO and COL validate that *A. guildingii* Schott corresponds to the description of *A. cordatum* (L.) Schott (Royal Botanic Gardens, Kew 2025; Catalogue of Life 2025; Acevedo-Rodríguez et al. 2012). However, the references that accept *A. guildingii* Schott use Fournet's Flora (Fournet 2002), which does not mention any collection in Martinique. Thus, the species recently recorded in Martinique and also accepted by the taxonomic references are:

Anthurium cordatum (L.) Schott sensu Howard (1979) is the most ecologically variable species, found in various vegetation types (mesophilous, hygrophilous, montane) between 200 and 1300 m altitude (Figure 3). Generally terrestrial and rarely epiphytic, it is caulescent to sub-caulescent. The inflorescence is short, erect, light green, and the fruits are red. The leaves are glossy green with a cordate venation (Howard 1979) (Figure 3).

Anthurium grandifolium (Jacq.) Kunth, from acaulescent to caulescent, is often epiphytic or climbing. It is distinguished by its large leaves (40–80 cm) and its drooping purple inflorescences, which can reach up to 1 m. The fruits are red to purple. The species tolerates various environments, from humid forests to drier zones (Howard 1979; Fournet 2002) (Figure 3).

Anthurium acaule (Jacq.) Schott, typically forming a “bird’s nest” is generally epiphytic and sometimes caulescent. It has lanceolate, leathery, dark green leaves and erect light green inflorescences. The fruits turn violet when mature. It prefers shaded, humid areas of mesophilous tropical forests (Howard 1979; Mayo 1982) (Figure 3).

Anthurium palmatum (Jacq.) Schott, caulescent and often epiphytic, is recognizable by its large, palmately lobed, dark green and leathery leaves. Its drooping inflorescences bear a pale yellow to pinkish spathe and red fruits. It is found in humid forests between 200 and 800 m altitude (Fournet 2002)(Figure 3).

Several classical revisions have grouped many *Anthurium* species under synonymy (Engler 1905; Mayo 1982; Howard 1979). However, more recent works have not systematically validated these synonymies and even propose enriched descriptions for certain taxa, notably *A. guildingii* Schott and *A. cordatum* (L.) Schott (Fournet 2002). Confusion persists between *A. cordatum* (L.) Schott and *A. guildingii* Schott, although *A. guildingii* Schott was accepted by Engler in 1905 (Engler 1905). Despite the synonymisation of this taxon by Howard in 1979, uncertainty remains about its precise delimitation, particularly regarding its distribution and morphological diagnosis (Fournet 2002; Howard 1979). Moreover, many photographs and herbarium specimens labeled *A. cordatum* (L.) Schott on platforms such as GBIF or Tropicos appear to actually correspond to *A. grandifolium* (Jacq.) Kunth (Global Biodiversity Information Facility 2025; Missouri Botanical Garden 2025). According to recent studies, despite the synonymisation of *A. cordatum* Engl. under *A. grandifolium* (Jacq.) Kunth, these specimens continue to be identified based on an incorrect description, revealing confusion on participatory platforms such as iNaturalist and PlantNet (Acevedo-Rodríguez et al. 2012; Fournet 2002; Howard 1979; iNaturalist 2025; Affouard et al. 2025). Recent observations on the island reveal morphologically distinct forms with no currently attached descriptions (Figure 4). Among



Figure 3: *Anthurium* species present in Martinique, respectively: Whole plant (A), Leaf (B), Spadix inflorescence (C). © Elian Vuillemot, © Benjamin Ferlay.



Figure 4: *Anthurium* Unidentified and undescribed species in Martinique. *A. sp. mtk.* (A), *A. cordatum* (*alt*) (B), *A. sp. (hybride)* (C). © Elian Vuillemot (A, B), © Benjamin Ferlay (C)

them, three specific cases noted by CBNMq botanists include: A potential hybrid between *A. acaule* and *A. grandifolium*, taxon not matching any existing description, observed in a single locality (*A. sp. (hybrid)*) ; A montane form of *A. cordatum*, collected and preserved in the MTK Herbarium, showing features specific to montane vegetation (*A. cordatum (alt)*) ; A primarily hygrophilic species with broad leaf blades clearly identified in CBNMq surveys. Its morphological characteristics do not match any known taxon, and no description is associated with it, possibly representing an undescribed taxa, here referred to as *A. sp. mtk.*. A significant morphological variability of *A. cordatum* has also been recently observed across different environments, ranging from xerophilous zones to montane vegetation in pitons. This variability, not mentioned in current descriptions, suggests either the existence of multiple taxonomic entities or strong environmental plasticity. Lastly, confusion appears to persist within the CBNMq as well. The main identification errors are as follows: individuals identified as *A. guildingii* actually match the description of *A. cordatum* (L.) Schott; those labeled as *A. cordatum* (L.) Schott correspond to *A. grandifolium*; and specimens identified as *A. grandifolium* correspond to *A. sp. mtk.* (Notes in ITMA). Numerous taxonomic sources are in disagreement, causing confusion in Martinique that could have repercussions on other islands where these taxa are also present. These discrepancies may significantly bias conservation efforts by misallocating resources to inappropriate taxonomic entities. This situation could even argue for the protection of populations rather than species, due to the absence of clear boundaries between taxa (Mace 2004; Rojas 1992). This highlights the need to revise the genus *Anthurium* in Martinique.

One increasingly used method to clarify the taxonomic and nomenclatural status of taxa is taximetry (Taylor 1966; Bisby et al. 1973; Prance et al. 1969). This quantitative method assigns numerical values to morphological or anatomical characteristics, allowing measurement of similarity between taxonomic units (Gidado et al. 2024; Awala et al. 2023; Ahmed et al. 2024). The method relies on multivariate statistical analyses to group taxa based on morphological similarities and is not suited to testing phylogenetic relationships (Zelditch et al. 1995). One major advantage of taximetry is its ability to improve botanical classifications, particularly in cases where descriptions or taxonomic resources are incomplete (Farouji et al. 2018). These methods also allow the study of morphologically similar taxa (Neto et al. 2019) and the identification of infrageneric or sectional groups using both qualitative and quantitative variables (Farouji et al. 2018). Interestingly, the application of taximetry to practical taxonomy was already envisaged before the widespread

use of modern computing tools. Prance (1969) emphasized that, even with increased computing power, the main task remains the rigorous collection of specimens and the precise coding of characters (Prance et al. 1969). Moreover, this method is an asset for the development of identification keys, as it highlights the most discriminating characters between groups, thus serving as an essential tool for identifying taxonomic groups (Awala et al. 2023).

We therefore asked whether morphometric analysis of the genus *Anthurium* in Martinique, combined with a study of its nomenclatural and taxonomic history, could clarify the current confusion and identify its origins. We hypothesize that these confusions may be related to the presence of at least one undescribed species on the island, as suggested by Howard and Fournet (Howard 1979; Fournet 2002). We also suggest significant intraspecific variability, particularly in *A. cordatum* (L.) Schott, which could lead to misidentifications. In this perspective, our hypothesis is that morphometric analysis is a relevant tool for delimiting taxonomic groups based on measurable morphological characters. Additionally, a critical examination of historical literature, taxonomic publications, and herbarium collections should help identify the main sources of confusion.

1.2 MATERIALS AND METHODS [1282 words]

Author Contributions

Sample collections were conducted by Benjamin Ferlay and the student. All analyses and the writing of this study were carried out by the student, with guidance from the supervisor.

Collaborative Project Background

This study is part of a broader collaborative project initiated during this internship, involving Guadeloupe, Martinique, potentially Dominica, and other islands of the Lesser Antilles, aiming to revise the genus *Anthurium*. The project includes several institutions: the Missouri Botanical Garden (MOBOT), the Jardin botanique de Lyon, the Conservatoire Botanique National Iles de Guadeloupe (CBNIG), and the Conservatoire Botanique National de Martinique (CBNMq), with the planned participation of Thomas Croat, a specialist in Aroideae. A collaboration has been established with the MOBOT Herbarium to carry out a genetic analysis. This analysis is based on the Angiosperm353 probe kit, which was specifically adapted for the genus *Anthurium* by Monica Carlsen-Krause (Carlsen-Krause in prep.). The kit includes approximately 200 genes from the Angiosperm353 set, along with around 800 nuclear genes specific to *Anthurium*. For this purpose, leaf samples of approximately 4 cm² were collected from young leaves of the studied species and dried in silica gel. All selected species correspond to those used in the morphometric analysis and are therefore referenced in one or more herbaria. In this study, genetic results are not presented or compared with statistical results due to sequencing and analysis delays. They will be the subject of a future independent publication.

The morphometric study focuses exclusively on native *Anthurium* species of Martinique, with confirmed occurrences and known localities, according to Howard's identification key (Howard 1979). Presence data were obtained from recent herbarium specimens (P, MPU, MO, MTK, GUA) (Thiers June 15, 2025), observations from citizen science platforms (iNaturalist, PlantNet), and the observations from the CBNMq. Specimens that could not be identified using existing keys were provisionally designated as *An-*

thurium sp., due to the lack of formal description. The species included in the study are: *A. cordatum*, *A. grandifolium*, *A. palmatum*, *A. acaule*, *A. sp. mtk.*, *A. cordatum (alt)*, and *A. sp. (hybrid)* (Figure 3, Figure 4). Conversely, *A. andraeanum* (non-native), *A. willdenowii*, and *A. dominicense* (without recent observations) were excluded. However, particular attention is given to these species in the field to ensure they are considered in the event of new discoveries. It should be noted that this is an unsupervised classification, and that the assigned species names will not be considered in the statistical analysis, as the analysis relies solely on morphological measurements, independently of the assigned names.

PILOT STUDY

To select the sampling sites, we relied on available literature and herbarium collections, using distribution data and associated ecological information. We also utilized data from the CBNMq, based on presence points or linear inventories. For *A. sp. mtk.*, *A. cordatum (alt)*, and the potential hybrid, no ecological or distribution data were available in the literature. We therefore used presence data from the CBNMq, supplemented with herbarium records for *A. sp. mtk.* and *A. cordatum (alt)*. Based on this analysis, we guided the field surveys for sample collection, primarily based on the knowledge and recommendations of the botanists from CBNMq. We conducted a stratified sampling by species in order to target the species we wanted to include in our analysis. Site selection was based on several criteria: the number of individuals observed per species to avoid impacting local populations, accessibility of the sites, ecological amplitude, and the maturity of the individuals encountered. At the end of the fieldwork, we selected 12 sites that met these criteria (Figure 5), representing the greatest possible variability among species and locations. In parallel, we tested 90 vegetative characters and 85 reproductive characters to determine which traits should be measured in situ and which were most relevant for statistical analysis (Figure 6, Table 1). The tested characters were based on standardized descriptions of the genus *Anthurium* (Croat and Bunting 1979). Additional traits were added from the revision of *Anthurium* in Mexico and Central America (Croat 1983). Finally, some traits deemed relevant were refined and added to complement the standardization.

SITES AND SAMPLING

Following the pilot study, the sites were selected according to phytosociological principles, i.e., in ecologically, floristically, and physiognomically homogeneous environments to encompass a broad range of variability. At each site, three individuals per target species were sampled, and multiple species could be collected at the same location. All individuals collected exhibited mature vegetative organs with reproductive parts at various stages of maturity. We distinguished between mature inflorescences (developed spadix, flowers open or not) and fruiting inflorescences (fertilized flowers). Individuals with multiple inflorescences were preferred in order to produce duplicates or even triplicates to be sent to associated herbaria. Individuals with multiple inflorescences were preferred in order to produce duplicates or even triplicates to be sent to international Herbaria (P, MO, NY).

Sampling involved collecting the entire individual for acaulescent species, and one meter of stem from the apical meristem for caulescent species, using a hand saw. Data related to the individuals sampled during the pilot study were recorded in a dedicated database, along with site-specific information. Recorded information includes: date, time, location, geographic coordinates, altitude, species, population size, population area, habitat type, humidity, sunlight, host species (if applicable), observers, and associated dominant

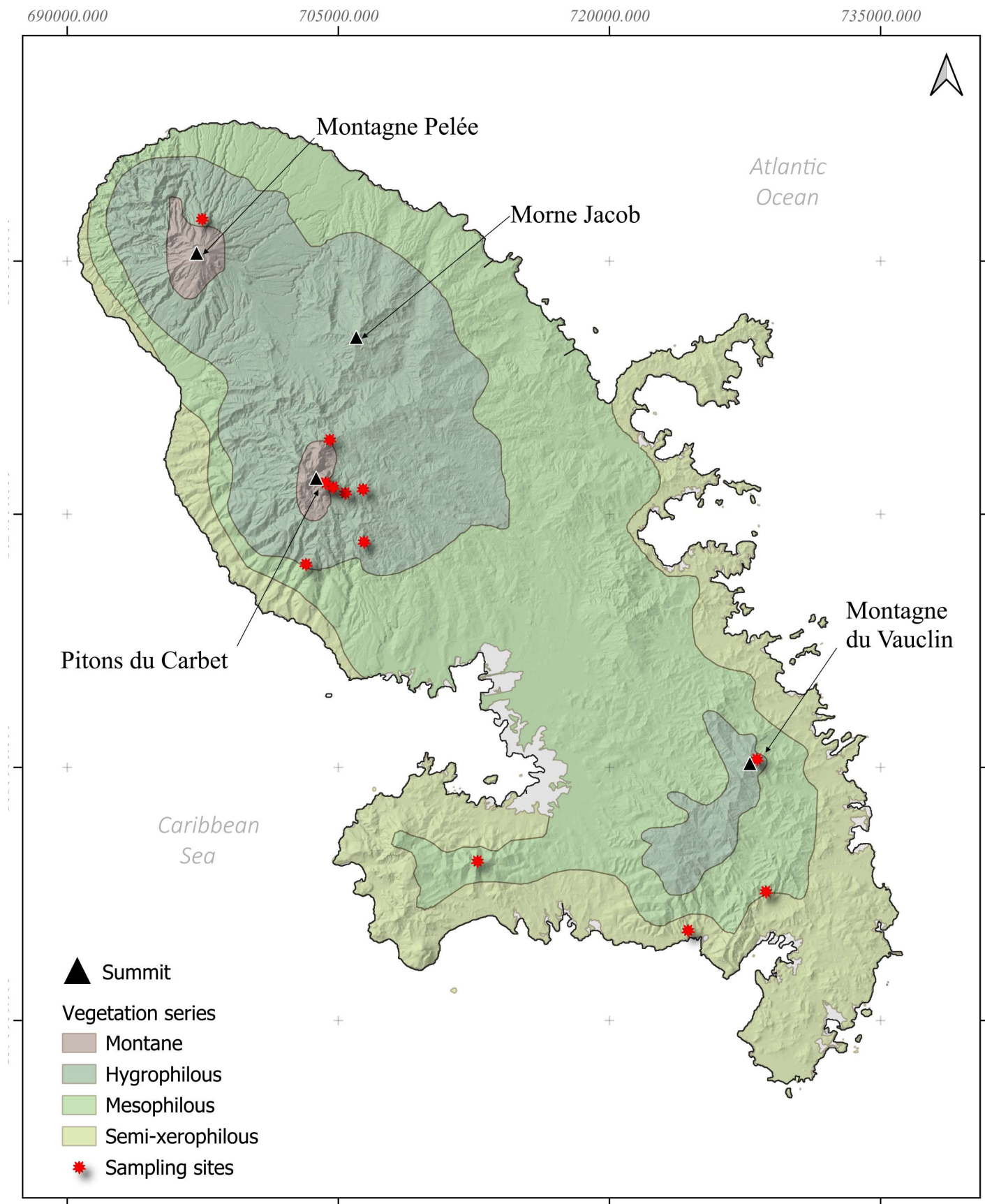


Figure 5: Sites selected for the collection of *Anthurium* specimens in Martinique. Background colors indicate vegetation types: montane (brown), hygrophilous (dark green), mesophilous (green), and semi-xerophilous (yellow). Data based on Portecop (1979) and transcribed by CBNMq (Portecop 1979). Major peaks are indicated by triangles. Pixel size: 25 × 25 m. © Elian Vuillemot

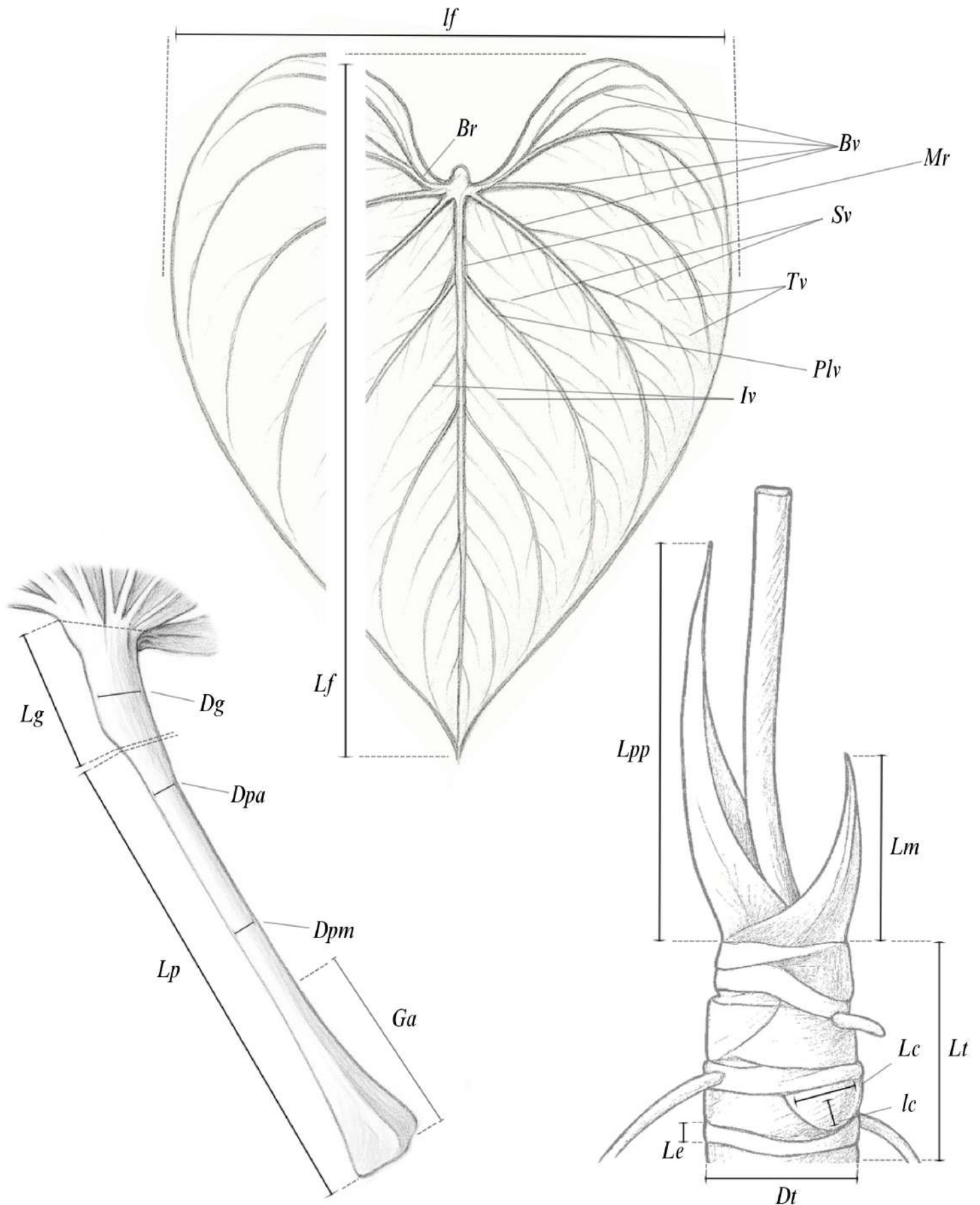


Figure 6: Vegetative characters measured on species of the genus *Anthurium* during the statistical analysis; only quantitative measurements and venation are illustrated. Leaf inspired by the standardized description of *Anthurium* (Croat and Bunting 1979). The drawings do not correspond to any specific species. Abbreviations are linked to (Table 1). © Elian Vuillemot.

species and habitat characteristics.

MEASUREMENTS AND HERBARIUM COLLECTION

The collected individuals were immediately analyzed in the laboratory within 24 hours, minimizing the effects of post-harvest drying and providing the best possible representation of *in situ* observations. Additional information on the individuals' observations after drying is included, if necessary. Measurements were taken using a meter, a micrometric ruler, a binocular loupe, and a microscope. Each specimen was dried and pressed in an drying oven at 50°C for one week, with press renewals to prevent decay. Specimens were also cataloged in the Herbarium MTK database, accessible under the genus *Anthurium*, with *Eliau Vuillemot* as the main collector (Conservatoire Botanique National de Martinique 2025).

MORPHOMETRIC ANALYSIS

A multivariate analysis was conducted to verify whether the current delimitation is correct. If not, this analysis helps to identify distinct morphological groups. The analysis includes all collected individuals, namely A total of 41 individuals were included in the study: 12 individuals of *A. cordatum*, including 3 from high-altitude sites; 9 individuals of *A. grandifolium*; 9 of *A. sp. mtk.*; 1 individual of *A. sp.* (hybrid); along with 3 individuals of *A. palmatum* and 4 of *A. acaule*. Since *A. acaule* and *A. palmatum* are well-defined species with distinct morphological characteristics that are distant from other taxa, their sample size was reduced. The number of individuals per species also varied due to material availability. In total, 45 variables were selected for the statistical analysis ((Table 1), Figure 6). We focused on vegetative traits, which are mainly used for delimiting *Anthurium* taxa (Croat 1983). Each quantitative trait was measured once or several times per specimen, and the average was then calculated. The resulting matrix was analyzed using R Studio (version 12.1).

Initially, specimens were compared without considering specific identification, allowing for the grouping of morphologically similar specimens. This partition was then compared to the recognized species. This approach is commonly used to clarify taxonomic boundaries and detect discriminative morphological traits (Faria et al. 2009; Hong-Wa et al. 2014; Kaplan et al. 2012). To quantify the morphological differences between specimens, pairwise comparisons were performed using a dissimilarity coefficient. The Gower dissimilarity index was selected, as it is particularly suited for data composed of heterogeneous variables, including binary, qualitative, and quantitative variables (Gower 1971). This method enabled the construction of a dissimilarity matrix for all specimen pairs. The `vegdist` function from the `cluster` package was used to calculate this matrix (Maechler et al. 2025a). To facilitate the application of subsequent statistical techniques and optimize result clarity, the matrix was transformed by square root into Euclidean distance, as recommended (Gower 1971). A transformation of the distance matrix into a coordinate system was performed using Principal Coordinates Analysis (PCoA) to visualize the relative position of specimens in a reduced space, thus facilitating the identification of groups. Since PCoA is based on a distance matrix, the measured variables do not directly contribute to the construction of the axes (Borcard et al. 2018). Consequently, their contributions are not explicitly provided (Borcard et al. 2018). The contributions of numerical variables were estimated by summing the absolute correlations with all axes (Σ), while those of categorical variables were assessed using the `envfit()` function from the `vegan` package, allowing their projection onto the PCoA and the analysis of their influence (Oksanen et al. 2001).

We used the `dudi.pco` function, from the `ade4` package (Dray et al. 2007). PCoA was preferred over Principal Component Analysis (PCA), as the latter is less suited for this type of analysis (Pierre et al. 2014). To determine the inertia associated with each axis, eigenvalues were calculated, and a plot was generated to identify the optimal number of axes to retain. In addition to PCoA, a discriminant analysis was conducted using decision trees, function `rpart` from the `rpart` package (Therneau et al. 1999), to identify the most discriminating morphological variables between clusters, and thus between species (De La Estrella et al. 2009; Slovak et al. 2012).

Finally, a hierarchical clustering analysis (HCA) was performed using the `hclust` function from the `stats` package Kartikeya Bolar 2019, on the dissimilarity matrix computed with the `daisy` function from the `cluster` package Maechler et al. 2025b. The most appropriate aggregation criterion was selected by comparing actual and projected distances using the Euclidean norm index and compared to the threshold value, in order to best represent the true distances Mérigot et al. 2010. The results were represented as a dendrogram, with branches reflecting dissimilarity between specimens. The leaves of the dendrogram were colored according to the PCoA, using the customized function `fColorLeaf` Pierre et al. 2014. The branches of the dendrogram were cut to form coherent clusters, and the validity of these groupings was verified using the Calinski-Harabasz index (calculated with the `intCriteria` function from the `clusterCrit` package (Desgraupes 2012).

Based on the results of the morphometric analysis, and to study the confusion surrounding *A. cordatum*, we will use data from different localities to determine if there is a gradient of morphological variability depending on the environment. We will also analyze which morphological traits are involved in this variation.

TAXONOMIC TREATMENT

Old descriptions were used to carry out the taxonomic revision within the genus *Anthurium*. For this, all available herbarium sheets of *Anthurium* were studied and re-identified using the most recent identification keys and descriptions Fournet 2002; Howard 1979. These references will be mentioned in the analysis. Information from descriptions based on specific type specimens was compared to other specimens in order to detect possible nomenclatural and taxonomic confusions. The entire GBIF database concerning specimens of the genus *Anthurium* Schott in Martinique was examined and supplemented by Herbaria collection such as Récolnat (P), the Muséum National d'Histoire Naturelle (P), the Missouri Botanical Garden (MO), the United States National Herbarium (US), and other external institutions (NY, K, B, etc.) Thiers June 15, 2025. Additional information such as the ecology, habitat, and geographic distribution of historical specimens were also recorded in order to compare the historical habitats of the species studied to their current distribution. Regarding the literature, bibliographic searches focused on horticultural journals, taxonomic works, local floras, and articles from scientific journals mentioning the identification of *Anthurium* species. Incomplete or questionable Herbarium specimens were analyzed with caution. For the genus revision and species descriptions, all rules followed are based on the International Code of Nomenclature for algae, fungi, and plants (ICN) Turland et al. 2018.

Table 1: Summary of vegetative characters measured on species of the genus *Anthurium* for statistical analysis. Variable types are indicated as follows: Ql. (qualitative), Ql. b. (binary qualitative), and Qt. (quantitative). The "Code" column shows the abbreviation used in Figure 6. The last column indicates whether the measurements were taken in situ. Variables removed due to multicollinearity with one or more other variables are marked with an asterisk

Tissues	Characters (Variables)	Character states (Modalities)	Variable type	Code	In situ
Root	Type	Clinging, nourishing, tuberous	Ql.	-	Yes
	Direction	Drooping, horizontal, upright	Ql.	-	Yes
	Density	Spacing, slightly, very compact	Ql.	-	Yes
Stem	Length*		Qt.	Lt	Yes
	Growth	Sub-, acaulescent, caulescent	Ql.	-	Yes
	Diameter*		Qt.	Dt	No
Internode	Length	-	Qt.	Le	No
	Shape	Uniform size, alternating	Ql.	-	No
Leaf scar	Horizontal length*		Qt.	Lc	No
	Vertical length		Qt.	lc	No
	Shape	Half-circle, triangular, intermediate, others	Ql.	-	No
Prophyll	Length		Qt.	Lpp	No
	Dorsal side	Prominent ridge, smooth, fimbriate	Ql.	-	No
Mesophyll	Length*		Qt.	Lm	No
	Dorsal side	Prominent ridge, smooth, fimbriate	Ql.	-	No
Leaf	Number		Qt.	-	Yes
	Young position	Erect, spreading, drooping	Ql.	-	No
	Aged position	Erect, spreading, drooping	Ql.	-	No
Lamina	Length		Qt.	Lf	No
	Width*		Qt.	lf	No
	Number of lobes		Qt.	-	No
	Widest part	Above, below or at petiole insertion	Ql.	-	No
	Shape	Croat and Bunting 1979 Fig.2	Ql.	-	No
	Outline	Membranous, normal	Ql.	-	No
	Margin	Revolute, flat	Ql.	-	No
	Adaxial aspect	Dull, shiny, pruinose, satiny	Ql.	-	No
	Abaxial aspect	Dull, shiny, pruinose, satiny	Ql.	-	No
	Punctuation	Yes, no	Ql. b.	-	No
Petiole	Length		Qt.	Lp	No
	Basal diameter*		Qt.	-	No
	Median diameter		Qt.	Dpm	No
	Apical diameter*		Qt.	Dpa	No
	Cross-sectional shape	Refer to Croat 1983	Ql.	-	No
	Channeled	Yes, no	Ql. b.	-	No
Sheath	Length		Qt.	Lga	No
Geniculum	Length		Qt.	Lg	No
	Median diameter*		Qt.	Dg	No
	Channeled	Yes, no	Ql. b.	-	No
Primary veins	Abaxial position	Sunken, flat, raised	Ql.	Mr	No
	Adaxial position	Sunken, flat, raised	Ql.	Mr	No
Primary lateral veins	Abaxial position	Sunken, flat, raised	Ql.	Plv	No
	Adaxial position	Sunken, flat, raised	Ql.	Plv	No
	Number of pairs		Qt.	-	No
Basal veins	Number		Qt.	Bv	No
Basal nerves	Length*		Qt.	-	No
	Trajectory	Ascending, descending, curved, straight	Ql.	-	No

1.3 RESULTS [1836 words]

MORPHOMETRIC ANALYSIS

We removed, on the one hand, the variables that were correlated with each other to avoid multicollinearity causing redundancy in the statistical analysis (Table 1), as well as the qualitative variables whose modalities were the same for all individuals, in order to limit unnecessary noise from these variables. Finally, the variables that we considered to have low reliability: all organ colors were removed, as this is subjective depending on the observer, the lighting, and varies even within a single individual. A part of the venation as well, for which the measured criteria changed between the first and last measurements due to a misidentification of the organs concerned. A total of 36 variables are used for the multivariate analysis (Table 1).

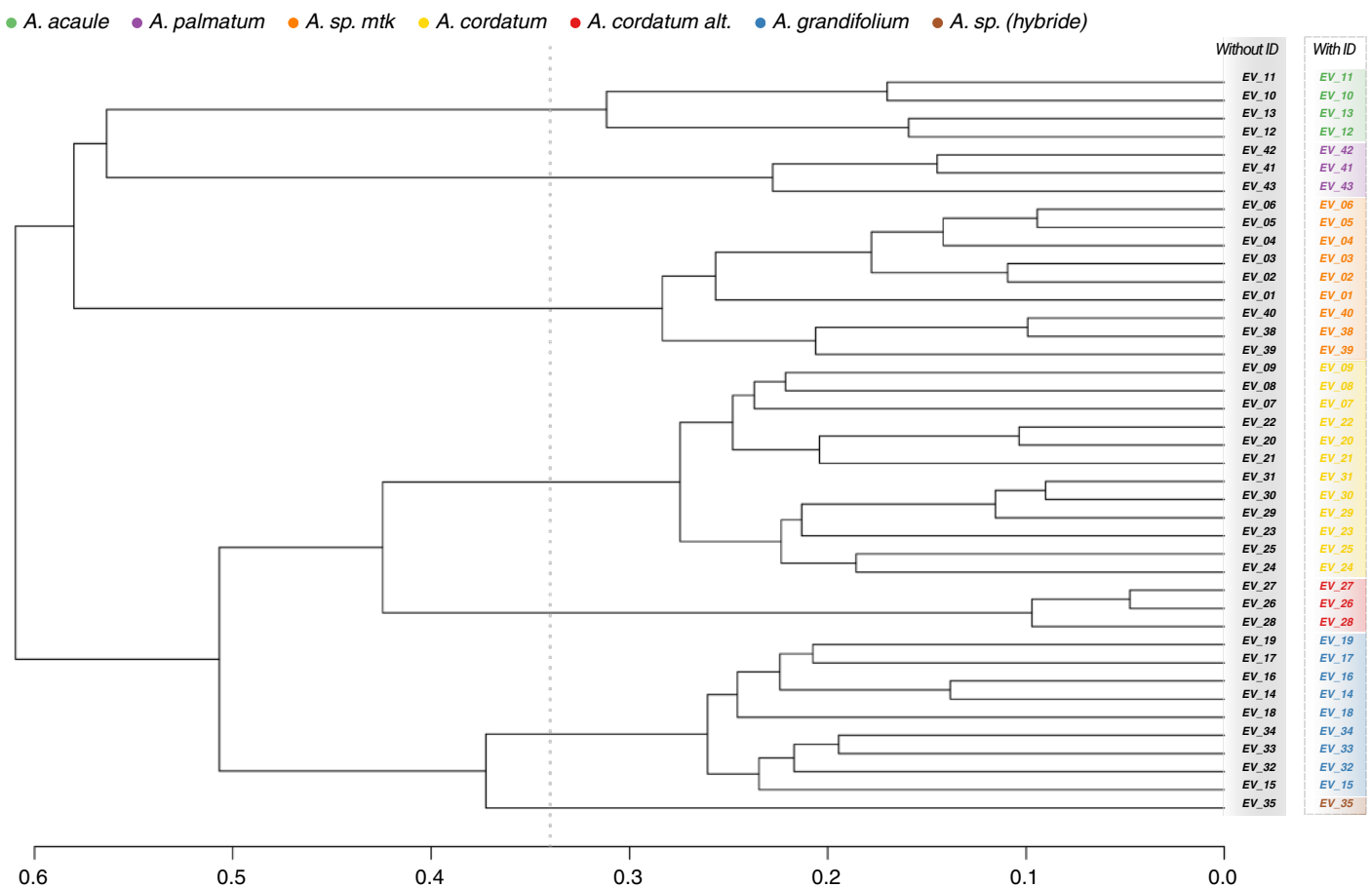


Figure 7: Dendrogram-based classification method for 41 individuals of the genus *Anthurium* from Martinique. The leaves respectively indicate: the collector's acronym – the collection number, and the color corresponds to the species identified prior to the measurements: *A. sp. mtk.* (orange), *A. palmatum* (purple), *A. cordatum* (yellow), *A. cordatum alt.* (red), *A. grandifolium* (blue), *A. sp. (hybride)* (brown), *A. acaule* (green). Distances were calculated using Gower dissimilarity, and the aggregation method used was average linkage.

The aggregation criterion selected for representing Gower distances was single linkage (Norm 2 = 3.35) (Appendix 3). However, the results will be interpreted with caution, as this value exceeds the defined threshold. Regarding the dendrogram, independently of species identification, the results show that certain individuals group together (Figure 7). The Calinski-Harabasz index suggests that retaining 7 clusters (149.03) is more consistent, although 6 clusters could also be suitable (143.80) (Appendix 4). The dendrogram clearly shows this separation of groups, mostly between 0.3 and 0.2 Gower dissimilarity (Figure 7).

We therefore retained 7 clusters for the rest of the analysis. When compared with the identification assigned to individuals, several groups can be distinguished. All species form well-identified groups, although *A. sp. (hybrid)* appears very close to *A. grandifolium*, and *A. cordatum alt.* to *A. cordatum* (Figure 7).

Based on the previous observation, we conducted a Principal Coordinates Analysis (PCoA), retaining 7 clusters (Figure 8). A total of 6 axes out of 40 were kept, accounting for 84.7% of the cumulative variance explained by the model. The relative position of all specimens could thus be visualized in a six-dimensional space. When representing 3 axes, 69% of the total inertia is explained: 35.6% by axis 1, 17.8% by the second, and 15.6% by the third axis. We also showed with the PCoA that no group overlaps (Figure 8). The assumption of homogeneity of within-group variances was not convincing ($P\text{-value[Betadisper]} = 1.45e-05$ ***), so the Permanova test results should be interpreted with this in mind. Although the PCoA and the dendrogram show that *A. sp. (hybrid)* forms a cluster isolated from the others, the Permanova showed that *A. sp. (hybrid)* is not significantly different from all other species (Table 2). It should be noted that we had only one individual for this taxon, and its internal variation is therefore null, unlike the other species. *A. palmatum* and *A. cordatum alt.* are distant in the graphical representations, but Permanova indicates they are not significantly different (Table 2); internal dispersions are also distant, respectively: 0.114 and 0.045. Other species appear to be significantly different according to Permanova (Table 2).

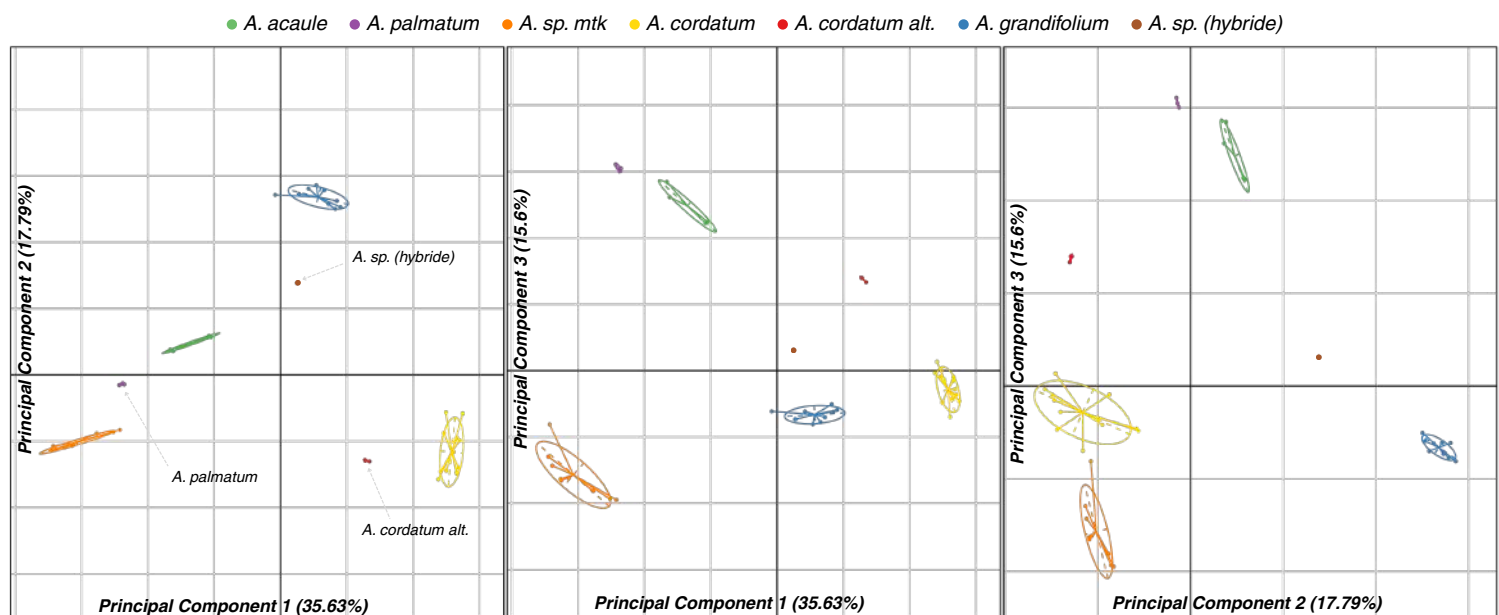


Figure 8: Principal Coordinates Analysis (PCoA) based on Gower distances for a sample of 41 individuals from the genus *Anthurium*. The colors correspond to the following species: *A. sp. mtk.* (orange), *A. palmatum* (purple), *A. cordatum* (yellow), *A. cordatum alt.* (red), *A. grandifolium* (blue), *A. sp. (hybrid)* (brown), *A. acaule* (green). The ellipses represent 95% confidence intervals. The percentages indicate the inertia explained by each principal coordinate.

The three most influential numerical variables in the distribution of the results are petiole length ($\Sigma = 2.06$), number of pairs of primary lateral veins ($\Sigma = 2.05$), and the length of the sheath ($\Sigma = 1.80$) (Table 4). Consequently, the correlated variables that were excluded are not directly included here, although some are correlated with these. The factorial variables that most strongly structure the distribution of the results statistically are root type ($R^2 = 0.95$ ***), root growth direction ($R^2 = 0.59$ ***), petiole cross-sectional shape ($R^2 = 0.70$ *** (Table 3).

Petiole length and cross-sectional shape constitute discriminating morphological characters (Table 3, Table 4, Figure 9). As the ANOVA assumptions are verified ($p\text{-value}[\text{Levene.Test}] = 0.096$; $p\text{-value}[\text{Shapiro-Test}] = 0.366$), a Tukey post-hoc analysis reveals that *A. sp. mtk*, *A. grandifolium*, *A. sp. (hybrid)*, and *A. palmatum* show no significant differences in petiole length (Figure 9). However, these taxa are clearly distinguishable by the shape of the leaf blade and that of the petiole cross-section. This observation also holds true for the other species.

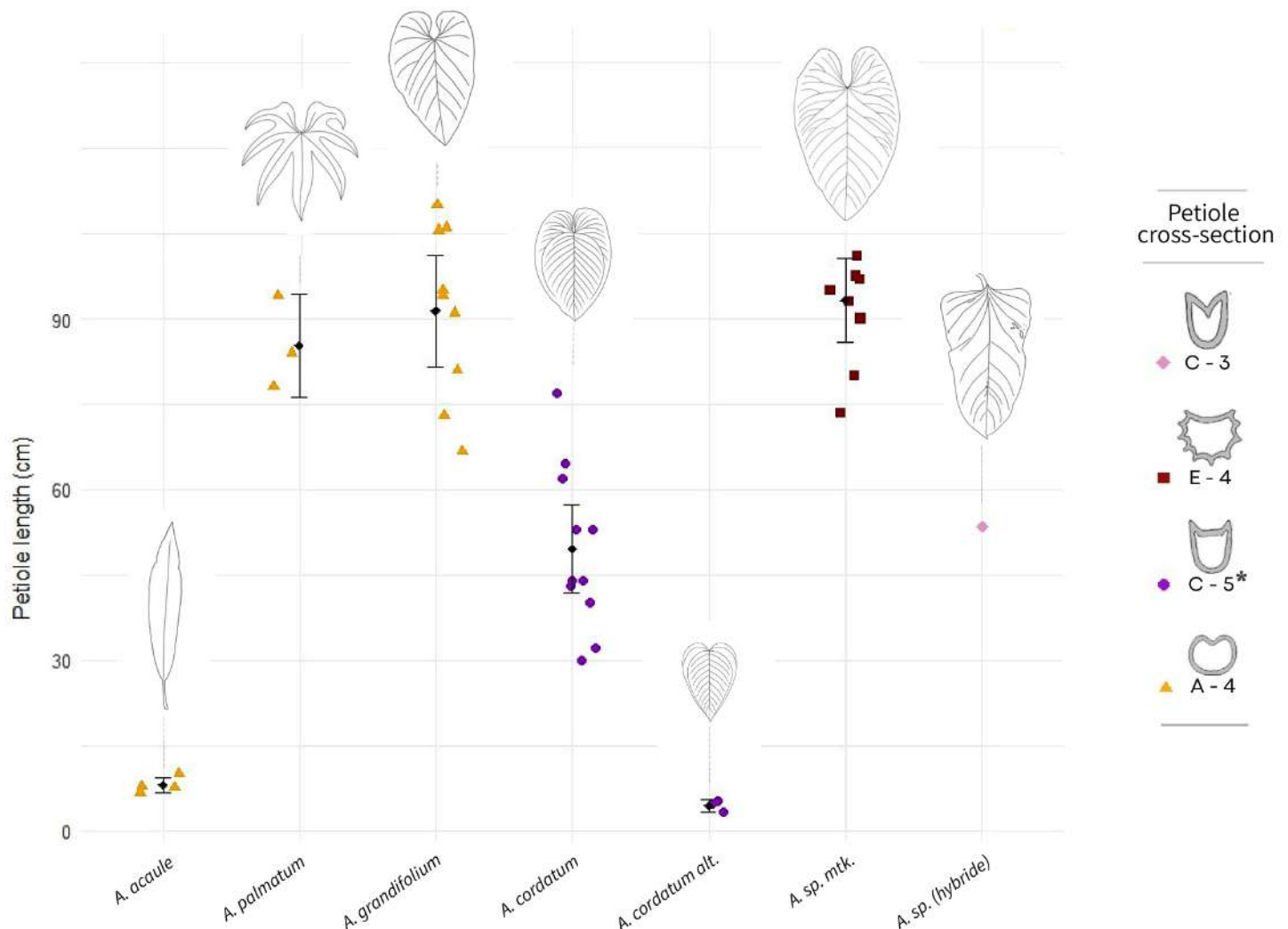


Figure 9: Petiole length (in cm) according to species of the genus *Anthurium* in Martinique. The dots represent sampled individuals; the shape of the dot indicates the category of petiole cross-section, described according to the standardized description of *Anthurium* (Croat and Bunting 1979). The leaf blade of each species is illustrated. The 95% confidence interval is shown ($\alpha = 0.05$). The homogeneous groups according to Tukey's test are: *A. sp. mtk*, *A. grandifolium*, *A. palmatum* ('a'); *A. sp. (hybrid)* ('ab'); *A. cordatum* ('b'); *A. acaule* and *A. cordatum.alt* ('c'). *The petiole shape of *A. cordatum* is variable: a ridge may appear on the lower arc.

TAXONOMIC TREATMENT

Concerning *Anthurium acaule* (Jacq.) Schott, we will not return to the species, as the work carried out by M. Simon Mayo from Kew provides a thorough study on *Anthurium acaule* and the West Indian 'Bird's Nest' Anthuriums (Mayo 1982). The taxonomic treatment is complete, however it does not mention Plumier's first description, whose original name was the polynomial *Arum foliis rigidis angustis et acuminatis* (Bot. Amer, vol. 3, 1689–1697). Likewise, for *Anthurium palmatum* (L.) Schott, the existing

studies seem clear and are consistent with the description (Fournet 2002; Croat and Carlsen 2013; Acevedo-Rodríguez et al. 2012). We consider that there is no taxonomic confusion for this taxon in Martinique.

Anthurium cordatum (L.) Schott, *Wiener Zeitschr. Kunst* 3: 828. 1829. **Basionym:** *Pothos cordata* L. *Sp. Pl.* ed. 2: 1373, 1763. **Type:** Guilding, L. [s.n.] Saint Vincent and the Grenadines (K000434039 and K000434040).

A. cordatum (L.) Schott was first mentioned under the name *Arum acinis amethystinis, flore parvo*, illustrated on plates 51 and 52 of Plumier (*Bot. Amer.*, vol. 3, 1689–1697). Linnaeus later proposed another polynomial name in the genus *Pothos*, under the name *Pothos foliis cordatis*, also known as *Pothos cordata* L., currently considered the basionym (*Sp. Pl.* ed. 2: 1373, 1763), referring to the illustration published in *Pl. Spe.*, vol. 4, p. 3, t.38, p. 205 (1756) by Burman and Plumier, which corresponds to plate 52 of Plumier's original manuscript. In 1829, Schott established a new combination with *Pothos cordata* L. becoming the basionym of *Anthurium cordatum* (L.) Schott (*Wiener Zeitschr. Kunst* 3: 828, 1829). It is noteworthy that no historical mention has been found of a change of epithet from *cordata* to *cordatum*. The typification of the species was made by G. Don, who designated as lectotype Plumier's illustration: "*Plum. ic. 26. t.38*" (*Hort. Brit.* ed. 3, p. 633, 1839). In 1857, Schott described several species: *A. guildingii* in Guadeloupe, *A. fallax* and *A. concinum* in Saint Vincent (*Oestr. B. Wehbl.*, 1857). In 1860, he revisited these descriptions and added a new one, *A. isertianum* for Guadeloupe, also reporting the presence of *A. guildingii* in Saint Vincent (*Prodr. Syst. A.*, 1860). In 1979, Howard placed in synonymy with *A. cordatum* (L.) Schott the four aforementioned species (Howard 1979).

Although Schott never mentioned these species in Martinique, several specimens were later identified as *A. guildingii* Schott (*CLF136101*; *Jérémie* 485; *Hahn* 353, 1080, 1244, 1427; *Privault* 272; *Sastre* 391, 6551; *Boubée* 6156; *Arnaud* 113; *Belanger* 111, 222, 365; *Rodriguez* 3618) or *A. isertianum* Schott (*Hahn* 1244; *Croat* 69755, 78387).

The individuals observed in Martinique match perfectly plate t.52 of Plumier (1689–1697) and its description. Herbarium specimens from Martinique, identified under the names later synonymized by Howard, actually correspond to *A. cordatum* (L.) Schott. However, individuals referred to here as *A. cordatum alt.*, collected in the col of Piton de l'Alma, had never been previously reported nor collected. They are characterized by an erect stem of 30–40 cm and very reduced organs (Figure X). Similar intermediate forms, observed during our surveys, had previously been collected on the Pitons du Carbet, identified at the time as *A. guildingii* Schott, at an altitude of 800–1000 m, and once described as "shrub" (*Boubée* 6156; *Sastre* 391, 6508).

Anthurium grandifolium (Jacq.) Kunth, *Enum. Pl.* 3: 77. 1841. **Basionym:** *Pothos grandifolia* Jacq. *Collect.* 4: 121. 1790; *Ic. Pl. Rar.* 3: t. 610. 1792. **Type:** Hahn M. [1317], Martinique

A. grandifolium (Jacq.) Kunth was initially described by Jacquin in 1791 under the name *Pothos grandifolia* (*Collectanea* 1791 Jacquin), found in Caracas, Venezuela. Kunth later established the new combination with *Anthurium grandifolium* in 1841 (*Enum. Pl.* III, p. 76). Jacquin's description seems to be based on the work of Plumier (*Plum. Amer.* 4-8- t. 63.), where the polynomial name *Arum amplis foliis cordiformibus, radice nodosa rubra* was proposed, with an illustration similar to *Icones* III, t. 610, Jacq. Regarding the red base of the stem mentioned in the works of Jacquin and Plumier, this trait has only been observed in the

mesophilic environments of the island, and not in the hygrophilous environments.

Additionally, Willdenow, in 1809, described *Pothos cordata* Willd. based on illustration 38 by Plumier (Plumier, 1689–1697). He also mentioned a herbarium specimen (Willdenow 3101, B-W03101-010), believing the illustration was made from “an individual with undeveloped spadices” (Enum. Pl. p. 168, 1809). Howard (1979) clearly states that this description and plate actually correspond to *A. grandifolium*. However, this same illustration subsequently led to many other descriptions, all now considered synonyms of *A. grandifolium* (Jacq.) Kunth, including *A. cordatum* (Willd.) Engler (Bot. Jahrb. Syst. 25: 368. 1898), *A. cordatum* (Willd.) G. Don. (Hort. Brit. 633. 1839), *A. cordifolium* Kunth (Enum. Pl. III. p. 76, 1841). Furthermore, the descriptions by Fournet (2002) and Howard (1979) both mention a note referring to a different form of this species, characterized by angular and striated petioles.

***Anthurium madininae* sp. nov. [auct. ined.], Martinique**

The taxon we have provisionally named *Anthurium* sp. mtk., which appears to correspond to the brief note mentioned by Fournet (2002) and Howard (1979), resembles a species (*t.* 53) first described by Plumier (Plumier, 1689–1697). The description under the polynomial name *Arum scandens amplissimo folio fructu longissimo* places the species at Cabesterre (Pitons du Carbet) in Martinique. It is characterized by its scandent, caulescent growth form, and by its very large leaves and inflorescences “*amplissimo folio fructu longissimo*”. Plumier mentions leaf blades reaching up to “6 feet” in length, which is consistent with the fact that *A. sp. mtk* is the only species with large blades reaching up to 135 cm. An incomplete illustration of the leaf accompanies the description and constitutes part of the protologue within the meaning of Article 38.7 of the Code. However, this mention by Plumier has never been cited in a valid binomial publication, not even by Linnaeus. No name has therefore been validly published according to Article 32.1, due to the lack of a proper description and rank indication. The absence of a later binomial combination precludes any retroactive validation (Art. 33.1). Some specimens (Hahn 1375, 354) may correspond to this taxon, but their condition or incomplete material renders identification difficult, especially given the lack of ecological data.

The present work proposes *Anthurium madininae* as the name for a new taxon (Art. 6 Note 1, Art. 23.3). A holotype is designated [Vuillemot 01, MTK], in accordance with the requirements of Art. 40.1. Since the validating description is not yet published, no author citation is indicated, in accordance with Art. 46.2 Note 1. We suggest a preliminary description in Appendix 5. The former pre-Linnaean polynomial name “*Arum scandens amplissimo folio fructu longissimo*” (Plumier 1689–1697) does not constitute a valid name (Art. 12.1, 12.2), as it predates the starting point of nomenclature in 1753. The historical illustration (Plate 53) (Plumier, 1689–1697) is designated as interpretative and reference material (Art. 9.4, 38.7).

***Anthurium xillobatum* nothosp. nov. [auct. ined.], Martinique**

What we have referred to as *A. sp.* (hybrid) is morphologically distinct from the other species on the island, notably by its venation and its erect, lance-shaped leaf blade lacking basal lobes. No mention or collection of any similar taxon has ever been made in Martinique. However, *A. selloum* C. Koch (Koch, Ind. Sem. Hort. Berol. App. 8. 1855), is described as a hybrid between *A. crenatum* (L.) Kunth and *A. cordatum*, endemic to the Virgin Islands (Acevedo et al. 2005). The herbarium descriptions and plates correspond to this description, except for the distribution range and parentage. *A. crenatum* is not present in Martinique

and is restricted to the Greater Antilles, but *A. acaule* also belongs to the section *Pachyneurium* (Mayo 1982), and *A. cordatum* (L.) Schott sensu Acevedo 2005 actually corresponds to *A. grandifolium*, based on the illustration and description mentioning a spadix up to 30 cm long and a cylindrical petiole (Acevedo et al. 2005).

Moreover, a recent field observation of *A. selloum* has been published, where the individual appears to be *A. crenatum* and *A. grandifolium* (iNaturalist 2025). It is worth noting that the individual found in Martinique is the only known representative to date, lithophytic in a mesophilic ravine environment, with nearby presence of *A. acaule* and *A. grandifolium*. The illustration by Saunders in *Refugium Botanicum* in 1869, identified as *A. macrophyllum* Schott (later synonymized with *A. selloum*), shows erect inflorescences, while the Martinican individual has horizontal to slightly oblique inflorescences. Other individuals with identical morphology have been observed on the island of Saint Lucia, where *A. grandifolium* and *A. crenatum* are both present, in addition to *A. hookeri* Kunth.

The present publication proposes *Anthurium ×illobatum* as a new nothospecies (Art. H.3.1). The proposed holotype for *Anthurium ×illobatum* is [Vuillemot 35, MTK] (Art. 40.1, Art. H.3.3). This hybrid is presumed to result from the parental species *A. acaule* and *A. grandifolium*, both present at the site where the individual was collected. In the absence of a validated name, the designation of the hybrid may be expressed in the form *Anthurium acaule* × *Anthurium grandifolium* (Art. H.3).

1.4 DISCUSSION [1754 words]

MORPHOMETRIC ANALYSIS

Multivariate analysis identified seven distinct groups, although the PERMANOVA was not significant. However, the observed heterogeneity of variances prevents a reliable interpretation of this test (Anderson and Walsh 2013), as it does not allow differentiation between differences related to centroids and those caused by internal dispersion (Warton et al. 2012). Thus, a significant PERMANOVA result may reflect internal variability rather than actual group separation (Anderson, Ellingsen, et al. 2006; Warton et al. 2012). Despite this limitation, the representation of the results revealed groups with strong dissimilarity. The combined use of the dendrogram and the PCoA reinforced this interpretation: the dendrogram highlights hierarchical relationships and clustering, while the PCoA, structured on several axes with high inertia, distinguishes the continuities and overlaps between groups (Legendre et al. 2012). Although based on visual analysis, this approach makes it possible to clearly delimit the identified groups.

Some limitations in the use of the Gower index were identified, particularly when applied to qualitative variables with numerous modalities, which may dilute relevant information (De Bello et al. 2021; D’Orazio 2021; D’Orazio 2024). It would be advisable for future analyses to use functions that treat the contribution of each variable uniformly, such as “*gawdis*” which adapts Gower distance (De Bello et al. 2021), to favor simple binary variables (presence/absence), which are more robust, or to code simple nominal variables using the Gower method in the same way as simple nominal data, in order to reflect intensity gradients (Gower 1971). Another suggested method is the use of a hierarchical system with primary and secondary characters, with the similarity of the secondary ones modulating the weighting of the primary ones when these characters are present (Williams 1969; Kendrick 1965; Proctor et al. 1963). This coding would allow for a better

consideration of morphological variations in inter-individual distances, and thus natural dependencies (i.e. if there is no stem, stem length and diameter variables cannot be measured) (Williams 1969; Kendrick 1965; Proctor et al. 1963). In this context, a correspondence factor analysis (CFA) restricted to qualitative variables, especially binary ones, would help identify the most discriminating characters for an identification key (Gourvenec 1994). In addition, a principal coordinates analysis (PCoA) based on a binary distance matrix provides a more neutral and balanced representation of the morphological structuring of individuals, which is particularly relevant in exploratory phases or in the absence of predefined groups (Borcard et al. 2018).

Furthermore, to make optimal use of Gower distance, it is important to pay attention to minimizing missing values and to how they will be handled (Gower 1971). It has been noted that this distance, although historically sensitive, has now been improved to handle missing values without compromising the Gower similarity coefficient (Caleb M. Brown et J. Arbour 2012; Kryszkiewicz 2023). In our case, missing values were rare and Gower distance was well suited for handling this data, particularly when applied to morphometrics (Dufour et al. 2009). No weighting was applied to the variables, so as not to bias the analysis with a priori assumptions about their contribution, and to allow the data structure to objectively highlight the most discriminating characters (Gower 1971). If the analysis were to be repeated on herbarium samples, care must be taken to ensure a sufficient number of specimens and to carefully select the characters to be measured so that they can be recorded without generating missing values. Finally, sample size also limited the scope of the analysis, particularly for taxa represented by a very small number of individuals, including the hybrid represented by a single individual (Xanthopoulos et al. 2013). Although unsupervised classification does not require a priori defined groups, a marked imbalance in group sizes may bias the results (Xanthopoulos et al. 2013). Indeed, the dissimilarity measures used in clustering methods, such as Gower distance or hierarchical analysis, are sensitive to data structure and may give disproportionate weight to the most represented groups (Legendre et al. 2012). Moreover, small samples tend to underestimate within-group variability, which may artificially reinforce the apparent separation between groups (Milligan 1980; Bandyopadhyay et al. 2013).

Nonetheless, morphometric analysis allowed the identification of *A. acaule*, *A. palmatum*, *A. cordatum*, *A. grandifolium*, as well as three other distinct groups. *A. acaule* and *A. palmatum* appear as well-delimited and easily identifiable taxa in Martinique, in accordance with recent studies and their clearly differentiated morphology from other species (Mayo 1982; Croat and Carlsen 2013). All Martinican taxa could be distinguished using only three morphological characters: leaf blade shape, petiole length, and petiole cross-sectional shape. These traits show promising potential for developing a future identification key.

TAXONOMY

For *A. cordatum*, although other names have been attributed in Martinique by various authors, we retain only the presence of *A. cordatum* (L.) Schott. This species nevertheless shows strong morphological variability depending on the environment, which we interpret as environmental plasticity rather than taxonomic differentiation. In dry and sunny environments, individuals display smaller, lighter, sometimes yellowish blades, with thinner and strongly grooved petioles. Conversely, in humid and shaded areas, the blades are very broad, dark green, and the petioles weakly grooved. Intermediate forms, notably with lanceolate or oval blades, may coexist in the same habitat or even on the same individual. Morphometric analysis suggests the existence of a distinct group for the high-altitude form of *A. cordatum*. The observed morphological variabil-

ity, which has caused taxonomic confusion with closely related species described by Schott (1857, 1860), could be explained on the one hand by strong environmental plasticity, and on the other hand by the presence of highly reduced individuals at higher elevations, especially in floral and foliar organs. Similar cases have been observed, notably in high-altitude herbaceous species, as well as in *Amorphophallus* and *Arisaema*, two genera also belonging to the Araceae, for which morphological reduction under altitudinal conditions has been documented (Ke et al. 2022; Oraphan Sungkajanttranon et al. 2019). We are currently working to determine whether this variation is of genetic origin (suggesting a subspecies or a distinct species) or a simple adaptive response of individuals to the environment. For this purpose, an *ex situ* cultivation of *A. cordatum* *alt.* has been initiated under the environmental conditions of *A. cordatum* to assess the plant's phenotypic plasticity. In parallel, ongoing genetic analysis will reveal its phylogenetic position.

This plasticity has already been reported for *A. intermedium*. Leaf shape, plant habit, venation, and dimensions of various parts (peduncle, petiole, midrib) are highly variable (Coelho et al. 2009). To avoid part of this bias, we suggest that in the future only mature individuals (i.e. with inflorescences) be studied, as during field sessions we observed great variability in juvenile individuals, which is commonly found in Angiosperms (Steenis 1955). *A. madininae* may resemble *A. cordatum*, *A. dominicense*, and *A. grandifolium* when young. *A. grandifolium* and *A. cordatum* can also be confused in hygrophilous vegetation.

We acknowledge that environmental variability is underrepresented in the present analysis. It is therefore necessary to include more precise variables illustrating this variability and to specifically test this hypothesis by taking into account an environmental gradient including hygrometry, altitude, and wind speed. Combining morphometric measurements with ecological data would be relevant, using Generalized Linear Mixed Models (GLMMs), which allow testing the effect of continuous or categorical environmental variables on morphology while accounting for the hierarchical structure of the data (Bolker et al. 2009). This type of model is particularly suited to evaluating the influence of the environment on morphometric traits and distinguishing fixed effects (environmental conditions) from random effects (site, individual, etc.) (Bolker et al. 2009). Furthermore, geometric morphometric analysis is a powerful and increasingly used method for quantifying shape and its variation in biological structures (Adams et al. 2004; Mitteroecker et al. 2009). Rather than simple linear measurements, it uses landmark coordinates to retain spatial and geometric information (Webster et al. 2010), making its application particularly interesting when combined with ecological variables. Finally, confusion persists around the taxa of Guadeloupe and Saint Vincent, leading to inter-island identification errors (Fournet 2002; Howard 1979). These confusions are partly due to the nomenclatures initially proposed by Schott and later synonymized with *A. cordatum* (L.) Schott by Howard, on which institutions generally agree.

Anthurium grandifolium constitutes a distinct group, although it is often confused with *A. cordatum*, largely due to a complex synonymy. Many descriptions seem to have caused confusion, including that of *A. cordatum* (Willd.) Engl. Confusion also stems, in part, from *Pothos cordata* Willdenow (1809). The herbarium plate (Willdenow 3101) morphologically corresponds to *A. grandifolium* (Jacq.) Kunth. Given that *A. grandifolium* can present a spadix as short as *A. cordatum* (L.) Schott during early development, Willdenow probably confused the two taxa. Many authors have since referred to this herbarium plate, which represents a juvenile individual of *A. grandifolium* with a young inflorescence that may resemble *A. cordatum*. This resemblance, combined with the morphological variability of *A. cordatum* Schott and the fact that both species have leaves described as cordiform, has contributed to the confusion. This initial error had significant

repercussions, causing confusion in taxonomic references, citizen science databases, and Herbaria, where more than half of the specimens of *A. grandifolium* are currently misidentified as *A. cordatum* (L.) Schott. Moreover, although *A. grandifolium* is described as a broad-leaved plant, confusion has also occurred with *A. madininae*, which has not been mentioned since 1697 and whose leaves are much larger. The foliage of *A. madininae* causes confusion, especially in young individuals of *A. grandifolium*. This confusion requires correction in databases, where *A. madininae* is currently mistakenly identified as *A. grandifolium*.

It would be interesting to study in more detail the different populations of *A. grandifolium* to verify whether the coloration of the stem and petiole base mentioned in early descriptions is indeed an environmental response, as we have observed. A thorough analysis including genetic data would allow determination of whether it constitutes a subspecies.

CONCLUSIONS AND PERSPECTIVES

Although from a statistical standpoint the analysis was not very conclusive regarding test validations, the study nonetheless highlighted both the current confusions—on the one hand, by identifying different individuals and groups of individuals according to their morphological characteristics, and on the other, through the taxonomic history of the genus *Anthurium* in Martinique. The taxonomic treatment proved more effective in identifying these confusions in our case, where many historical works have already examined previous descriptions; thus, it is easier to trace the taxonomic history and conclude on the origin of these confusions. Conversely, morphometric or even anato-morphometric analysis has already proven effective in taxon delimitation (Mutanen et al. 2007; Ospina-Garcés et al. 2022; Mayo, Xanthos, et al. 2023; Pierre et al. 2014), but it is often more powerful when combined with genetic analysis, particularly to detect cryptic taxa and to identify erroneous taxonomic descriptions linked to environmental plasticity (Rudoy et al. 2022; Karbstein et al. 2023).

We are currently working on a revision of the genus at a broader scale, including neighboring islands, in order to clarify these uncertainties and enable an update of taxonomic references as well as participatory databases. Formally naming an entity as a taxon allows for the application of international standards in terms of population monitoring, threat assessments, and conservation status (Mace 2004). This enables scientific monitoring at multiple scales, both local and global, whether to determine their ecological role, distribution, density, or dynamics. These criteria play a crucial role in political decisions, particularly regarding the allocation of funding (Mace 2004). A global revision of the genus *Anthurium* in the Antilles would thus allow for better understanding and use of taxa by conservation organizations, for which standardized identification is essential to the coherence and effectiveness of management actions.

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1.5 APPENDICES

Table 2: PERMANOVA results (9999 permutations) based on Gower distances among species of the genus *Anthurium*. *p*-values are annotated according to significance levels: * < 0.05; ** < 0.01; *** < 0.001.

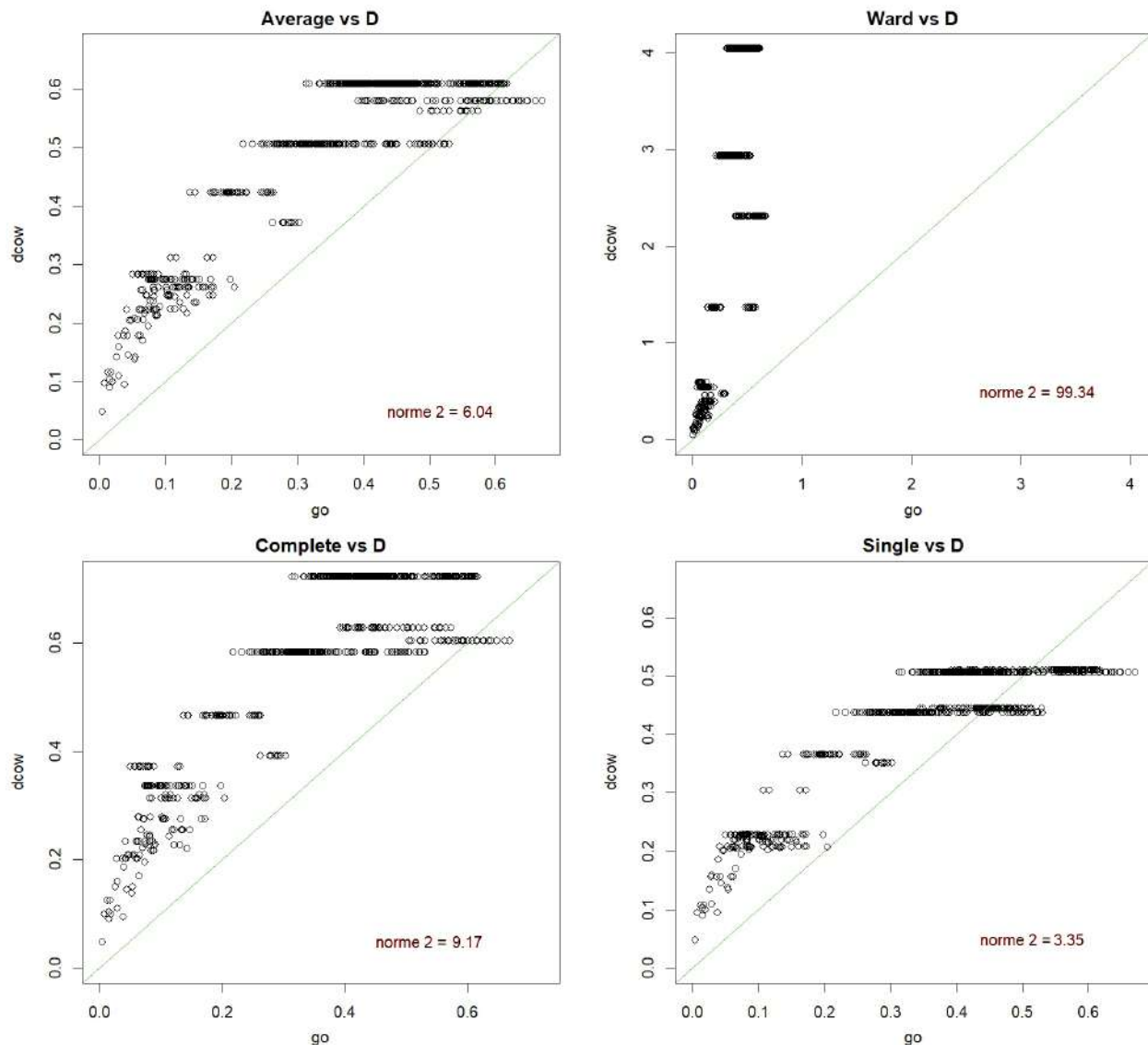
Species	<i>A. cordatum</i>	<i>A. cordatum</i> <i>alt.</i>	<i>A. acaule</i>	<i>A.</i> <i>grandifolium</i>	<i>A. sp.</i> (Hybride)	<i>A. palmatum</i>
<i>A. cordatum alt.</i>	0.00570**	—				
<i>A. acaule</i>	0.00053***	0.04286*	—			
<i>A. grandifolium</i>	0.00053***	0.00753**	0.00570**	—		
<i>A. sp. (Hybride)</i>	0.10542	0.25000	0.22105	0.11853	—	
<i>A. palmatum</i>	0.00753**	0.11853	0.04286*	0.00753**	0.25000	—
<i>A. sp. mtk</i>	0.00053***	0.00753**	0.00570**	0.00053***	0.11853	0.00753**

Table 3: Goodness of fit (R^2) between individual categorical traits and species positions in the PCoA ordination. *p*-values are annotated according to significance levels: * < 0.05; ** < 0.01; *** < 0.001.

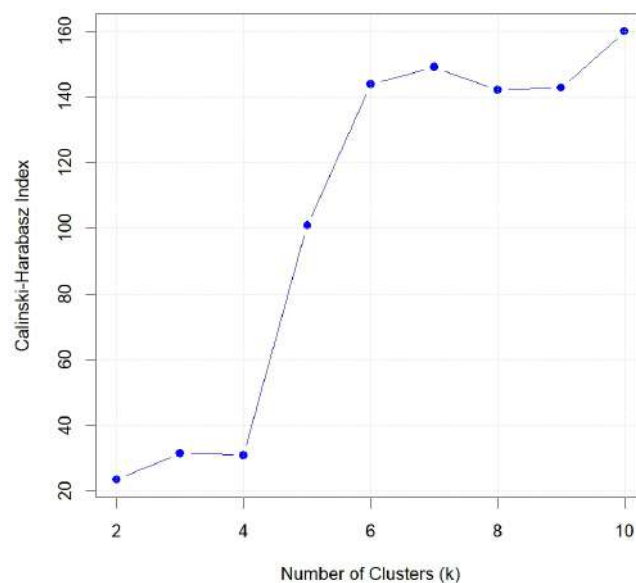
Tissue	Character	R^2	<i>p</i> -value
Root	Type	0.8295	0.001***
Root	Density	0.8179	0.001***
Petiole	Cross-sectional shape	0.7326	0.001***
Root	Direction	0.6557	0.001***
Primary veins	Adaxial position	0.5711	0.001***
Leaf scar	Shape	0.5700	0.001***
Lamina	Shape	0.4798	0.001***
Primary veins	Adaxial position	0.4544	0.001***
Prophyll	Dorsal side	0.4146	0.001***

Table 4: Global contribution (Σ) of quantitative variables to the PCoA ordination, calculated as the sum of absolute Pearson correlations between each variable and the six retained axes.

Tissue	Character	Contribution
Petiole	Length	2.058841
Primary lateral veins	Number of pairs	2.048017
Sheath	Length	1.799884
Petiole	Median diameter	1.798557
Lamina	Length	1.797166
Basal veins	Number	1.782683
Leaf scar	Vertical length	1.765661
Geniculum	Length	1.656919
Internode	Length	1.632152



Appendix 3 : Gower dissimilarity (go) as a function of cophenetic distance (dcow) according to aggregation criteria: average linkage, complete linkage, single linkage, and U-Ward linkage. The Euclidean norm values are given for each.



Appendix 4 : Calinski-Harabasz index applied to Gower dissimilarity on the morphometric dataset of *Anthurium*



Appendix 5 : Spathe (A); Spathe insertion (B–C); Whole inflorescence (D), Inflorescence (H); Peduncle cross-section, basal (E), median (F), apical (G); Flowers (I), Dissected flowers (J), Stamens (K); Fruits (V); Stem and cataphylls (M); Leaves adaxial surface and punctation (N), abaxial surface (O); Venation (P–Q); Blade insertion (L–R); Petiole base (S); Entire petiole (T); Petiole cross-section basal, median, apical (U).

Anthurium madininae [auct. ined.], sp. nov.

Type: MARTINIQUE, Morne Césaire, 479 m, [14.6746038, -61.0842225], Date: 11/04/2025, Phenology: Flowering, Elian Vuillemot, Benjamin Ferlay [holotype: MTK (barcode undefined); isotype: GUA (barcode undefined), MO (barcode undefined)].

Diagnosis: *Anthurium madininae* is characterized by its large mature size, climbing habit, strongly channeled petioles, leaf blades averaging 1.5 times larger than those of *A. grandifolium*, very glossy slightly bullate green appearance, longer acuminate apex, and 8 pairs of basal veins. Spadix dark purple, curved upward, with short stamens; fruit yellow.

Description: Juvenile plant terrestrial or slightly lianescent in search of support, then becoming climbing epiphytic and caulescent, usually ascending from the ground. Mature plant reaching 5–8 m in height on its support. Monopodial growth, acrotonic branching when apex is cut. **Stem** diameter (30.3–) 35.4–51.55 (–57.08) cm. Short **internodes** variable in length (4.79–) 5.41–9.25 (–10.56) mm. **Roots** moderately sparse, for anchorage, horizontally spreading, dark brown to dark green, red when young. **Leaf scars** semi-circular (33.15–) 37.13–48.04 (–52.38) cm × (23.5–) 30.74–42.85 (–49.19) cm. **Prophyll** up to (9–) 10–17.2 (–20.3) cm long, membranous, light green, quickly turning light to dark brown, caducous after two cataphylls, marcescent on 2–3 nodes, then caducous. Veining faint, apex acuminate and generally mucronate, dorsal rib with two prominent slightly fimbriate crests. **Mesophyll** (20.5–) 22.5–33.5 (–40.3) cm, membranous, light green, quickly decomposing with brownish venation; same caducity pattern as prophylls. **Petiole** (73.5–) 90–97.5 (–112) cm long; median diameter (11.17–) 13.4–21.68 (–29) mm. **Transversal shape** E-4 throughout, medium green, matte, densely punctate with oxalate points, channeled. **Geniculum** (37.59–) 49.7–59.49 (–67.03) cm long, diameter (14.48–) 16.11–24.79 (–25.37) mm, slightly paler and thicker than petiole, same shape. **Sheath** length (13.5–) 14.5–25.5 (–28.5) cm. Often a silvery surface at petiole and cataphyll base. **Leaves** erect when young, spreading with age, simple, symmetrical. **Blade** shape from oval 3:2 to triangular-oval 2:1; length (78–) 88–121 (–133.5) cm; widest below insertion; width (46.5–) 54.5–68 (–76.5) cm. 1.5× longer than wide, slightly longer than petiole, sub-pendent. Apex slightly acuminate, base deeply lobed, spatulate to obovate, membranous, sometimes slightly bullate; adaxial dark green and glossy, abaxial paler and matte to semi-glossy. Notable punctation from calcium oxalate secretory cells adaxially. **Margin** smooth, slightly revolute, translucent and distinct from blade. **Venation** coriaceous and brittle. **Primary veins** prominent on both surfaces, dark green adaxially, lighter abaxially. **Lateral primary veins** (7–) 8–10 (–10) per side, straight, descending

from midrib. **Interprimary veins** (1–) 1–2 (–3) per side. **Basal veins** (7–) 8 (–9) per side, convex, fused over (29.55–) 48.89–77.09 (–88.81) mm, ascending, glabrous. **Secondary veins** generally 1 per lateral vein and 2–3 per basal vein; yellow-green; convex, raised adaxially. Tertiary veins green, slightly raised adaxially. **Fine reticulate** veins visible, flat. **Collective veins** present along margin and internally from basal and lateral primaries.

Inflorescence obliquely ascending; apex curved upward from early flowering. **Peduncle** 40–50 cm long; diameter (3.86–) 4.32–14.47 (–26.17) mm; grooved and randomly twisted. **Spathe** spreading; light green, matte above, glossy below, visible veining, long-acuminate and revolute apex; size (18.5–) 23.4–34 (–37) cm × (2.3–) 2.5–6.44 (–7.06) cm, widest at insertion, pendant at 30°, persistent to fruiting. **Spadix** tapered (Fig. 9-B, [Croat, 1979]), dark purple turning reddish at anthesis, matte to semi-glossy; (17.3–) 30–49.3 (–56.5) cm long; diameter (5.74–) 10.44–14.12 (–16.3) mm; slightly stipitate: posterior (3.83–) 7.37–12.15 (–18.94) mm, anterior (5.91–) 9.91–15.62 (–20.61) mm; **Stipe** light green, glossy. **Flowers** tetramerous, purple to deep red when aging, (30–) 33–56 (–69) per spiral; diamond-shaped arrangement, slightly undulate; length (2.5–) 2.79–3.76 (–4.45) mm along spiral and (2.99–) 3.28–3.55 (–4.18) mm perpendicular. **Tepals** (4), outer margin convex acute, internal margin concave; lateral tepal length (3.21–) 3.51–3.83 (–4.12) mm; anterior/posterior (1.34–) 1.58–1.76 (–2.13) mm. **Stamens** (4), barely emergent, ascending from base forming shield. **Anthers** yellow to orange with age, apicifixed, non-exposed filament, extrorse, dehiscing longitudinally; parallel to gynoecium, becoming orthogonal. Raphide cells present at theca apex. Anther width (0.57–) 0.66–0.82 (–1.75) mm; length (0.83–) 1–1.17 (–1.29) mm after anthesis. Filament (0.98–) 1.12–1.5 (–4.3) mm wide; length (0.97–) 2.97–3.69 (–3.88) mm. Pollen globular, yellow. **Pistil** pyramidal; apex purple, cruciform. **Stigma** scarcely visible, rounded, violet-red apex. **Fruit** yellow, slightly acuminate, brown-purple apex.

Distribution, ecology, and phenology: Pitons du Carbet, Morne Jacob, and around Montagne Pelée. Occupies hygrophilous forest from 450–1000 m elevation. Generally in dense sciaphilous forest, occasionally in semi-heliophilous habitats along ridge crests. Flowering: March–June; Anthesis: April–June; Fruiting: September.

Conservation status: The species is mostly present across the island in suitable habitat. No specific conservation status assigned to date. Status to be assessed under IUCN criteria.

Etymology: The species is named in reference to “Madinina,” the name given by Amerindian peoples (Arawaks or Caribs) to the island of Martinique prior to European arrival, meaning “island of flowers.”

1.6 ABSTRACT

FRENCH

Le genre *Anthurium*, l'un des plus diversifiés de la famille des Araceae, demeure encore peu documenté dans les Petites Antilles, et notamment en Martinique. Cette étude vise à clarifier l'identification et la délimitation des espèces indigènes d'*Anthurium* présentes sur l'île, en combinant une approche morphométrique rigoureuse à une révision taxonomique critique fondée sur les descriptions historiques et les collections d'herbiers. Sept entités morphologiquement distinctes ont été ciblées : *A. cordatum*, *A. grandifolium*, *A. palmatum*, *A. acaule*, une forme altitudinale de *A. cordatum*, un hybride putatif (*A. ×illobatum*), et un taxon non décrit nommé provisoirement *A. madininae*. Un protocole de mesure standardisé a été appliqué à 41 individus répartis sur 12 sites représentatifs. Après sélection et élimination des variables corrélées ou peu fiables, 36 caractères morphologiques ont été intégrés à des analyses multivariées non supervisées (PCoA, dendrogrammes, arbres de classification). Les résultats ont permis de mettre en évidence sept groupes bien différenciés, malgré les limites statistiques dues à l'hétérogénéité des variances et à la faible taille de certains échantillons. Trois caractères se sont révélés discriminants pour identifier les entités : la forme du limbe, la longueur du pétiole et la section du pétiole. Une forte plasticité phénotypique a été observée chez *A. cordatum*, notamment dans les milieux montagnards, soulevant l'hypothèse d'une adaptation environnementale ou d'une divergence génétique, actuellement testée par culture ex situ et analyses moléculaires. La révision des descriptions anciennes (Plumier, Willdenow, Schott) révèle une importante confusion historique entre *A. grandifolium* et *A. cordatum*, amplifiée par la variabilité morphologique intra-spécifique et par des erreurs dans les bases de données actuelles. Deux nouveaux taxons sont proposés ici : *A. madininae*, espèce probablement endémique de Martinique, et *A. ×illobatum*, issu de l'hybridation entre *A. acaule* et *A. grandifolium*. Cette étude souligne l'importance d'une approche intégrative combinant morphométrie, écologie, nomenclature et phylogénie. Elle constitue un socle indispensable à une révision du genre *Anthurium* dans les Petites Antilles, avec des implications directes pour la conservation, la mise à jour des bases floristiques, et la mise en œuvre de politiques de gestion adaptées à la biodiversité insulaire.

Mots-clés : *Taxonomie, Nomenclature, Taximétrie, Analyse multivariée, Anthurium, Martinique*

ENGLISH

The genus *Anthurium*, one of the most diverse within the Araceae family, remains poorly documented in the Lesser Antilles, and particularly in Martinique. This study aims to clarify the identification and delimitation of native *Anthurium* species on the island by combining a rigorous morphometric approach with a critical taxonomic revision based on historical descriptions and herbarium collections. Seven morphologically distinct entities were targeted: *A. cordatum*, *A. grandifolium*, *A. palmatum*, *A. acaule*, a high-altitude form of *A. cordatum*, a putative hybrid (*A. ×illobatum*), and an undescribed taxon provisionally named *A. madininae*. A standardized measurement protocol was applied to 41 individuals across 12 representative sites. After filtering out correlated or unreliable variables, 36 morphological traits were included in unsupervised multivariate analyses (PCoA, dendrograms, classification trees). The results revealed seven well-differentiated groups, despite statistical limitations due to variance heterogeneity and small sample sizes. Three traits proved particularly effective in distinguishing entities: blade shape, petiole length, and petiole cross-sectional profile. Marked phenotypic plasticity was observed in *A. cordatum*, especially in montane environments, raising the hypothesis of environmental adaptation or genetic divergence, currently being tested through ex situ cultivation and molecular analysis. A review of historical descriptions (Plumier, Willdenow, Schott) highlights significant taxonomic confusion between *A. grandifolium* and *A. cordatum*, exacerbated by intraspecific morphological variability and errors in current databases. Two new taxa are proposed: *A. madininae*, likely endemic to Martinique, and *A. ×illobatum*, resulting from hybridization between *A. acaule* and *A. grandifolium*. This study underscores the importance of an integrative approach combining morphometrics, ecology, nomenclature, and phylogeny. It forms a foundational step toward a comprehensive revision of the *Anthurium* genus in the Lesser Antilles, with direct implications for conservation, updates to floristic databases, and the implementation of management policies adapted to island biodiversity.

Keywords: *Taxonomy, Nomenclature, Taximetry, Multivariate, Anthurium, Martinique*

1.7 GLOSSARY

Term	Definition
Acaulescent / Caulescent	Without / with a visible stem
Calco-alkaline	Volcanic rock rich in alkalis, typical of subduction zones
Cataphyll	Leaf with a reduced blade
Cordate	Heart-shaped
Diagnostic morphology	Set of visible traits used to identify a taxon
Endemism	Natural occurrence of a species exclusively in a specific region
Epiphyte	Plant growing on another without parasitizing it
Hybrid species	Individual resulting from a cross between two different species
Infrageneric	Belonging to a subgroup within a genus
Intraspecific	Within a single species
Geniculum	Swelling between the petiole and the leaf blade
Herbarium	Collection of dried plant specimens used for reference
Hemiepiphyte	Plant that starts life as an epiphyte and later reaches the ground
Homoplasy	Evolutionary similarity not inherited from a common ancestor
Hygrophilous / Hygrophytic	Adapted to very humid environments
Lithophyte	Plant growing on rocks
Magmatism	Processes related to the formation and movement of magma
Mesophile / Mesophytic	Adapted to moderately humid environments
Mesophyll	Cataphylls appearing after the prophyll
Monophyletic	Group of organisms sharing a single common ancestor, including all descendants
Morphometrics	Quantitative study of the forms and structures of organisms
Multivariate analysis	Statistical technique handling multiple variables simultaneously
Nomenclature	Rules for naming organisms according to a scientific code
Petiole	Stem connecting the leaf to the main stem
Phyllotaxy (spiral)	Leaf arrangement in a spiral around the stem
Plasticity (environmental)	Ability of an organism to change morphology based on environment
Polyphyletic	Group that does not share a direct common ancestor
Prophyll	First cataphyll
Rheophyte	Plant adapted to fast-flowing riverbanks
Spadix	Fleshy axis bearing small flowers, typical of Araceae
Spathe	Colored bract surrounding the inflorescence
Subduction	Sinking of one tectonic plate beneath another
Sympodial	Growth where the main stem is replaced by a lateral shoot

Synapomorphy	Shared trait inherited from a common ancestor
Synonymy	Situation where several scientific names refer to the same taxon
Taxon (pl. taxa)	Group of organisms classified in biology
Taximetric analysis	Statistical method to compare biological groups using measurable traits
Taxonomic / Taxonomy	Science of classifying living organisms into hierarchical categories
Tepals	Floral parts when petals and sepals are identical
Tholeiitic	Potassium-poor volcanic rock, typical of mid-ocean ridges
Xerophile / Xerophytic	Adapted to dry environments
